

FILTER DESIGN TEST CASE

Coupled Resonator Filter - WiFi 2.4 GHz for Public Safety Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P2 - High

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Coupled Resonator Filter filter targeting WiFi 2.4 GHz for Public Safety Band Filter. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 3499.0 MHz and bandwidth of 296.9 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Public Safety Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Coupled Resonator Filter
- Frequency Band: WiFi 2.4 GHz
- Application: Public Safety Band Filter
- Substrate: 128° YX LiNbO3
- Package: SiP (System in Package)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Osaka Design Center

This specification applies to all variants of the Coupled Resonator Filter design targeting WiFi 2.4 GHz. Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3499.0 MHz
Bandwidth (3 dB):	296.9 MHz
Insertion Loss Target:	<= 1.2 dB
Return Loss Target:	>= 21.0 dB
Out-of-Band Rejection:	>= 31.0 dB
Isolation (Duplexer):	>= 27.0 dB
Q Factor:	4874
Temperature Coefficient:	-6.9 ppm/°C
Die Size:	1.3 x 1.8 mm
Wafer Size:	8-inch
Number of Resonators:	7
Electrode Material:	W

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the Coupled Resonator Filter template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 3499.0 MHz, BW 296.9 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Keysight N9041B UXA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 10497 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.2 dB
- Return loss in passband shall be >= 21.0 dB
- Out-of-band rejection at +/- 445 MHz offset >= 31.0 dB
- Group delay variation across passband shall be <= 9 ns
- Temperature drift over -40°C to +85°C shall be <= 3.0 MHz
- Power handling shall be >= +27 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 89%

6. TEST RESULTS

Measurement Date: 2024-04-25
Devices Tested: 62
Insertion Loss (meas): 0.9 dB
Return Loss (meas): 22.7 dB
Rejection (meas): 33.7 dB
Isolation (meas): 31.7 dB
Q Factor (meas): 4874
Group Delay Variation: 2.6 ns
Power Handling: +26 dBm
Die Yield: 97.9%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The Coupled Resonator Filter design demonstrated below-target performance for WiFi 2.4 GHz.
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Group delay flatness improved compared to previous revision.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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